Predicting **wildfire** induced changes to runoff: A **review** and synthesis of modeling approaches

D Partington, M Thyer, M Shanafield… - Wiley …, 2022 - Wiley Online Library

… of different **model** types, improvements in predicting **wildfire** induced hydrological impacts
… The ET **dynamics** post-**wildfire** is also dependent on the regeneration strategies of different …

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A systematic **review** of applications of machine learning techniques for **wildfire** management decision support

K Bot, JG Borges - Inventions, 2022 - mdpi.com

… The 21 papers addressing this **wildfire** cycle management … : **wildfire** detection, **wildfire spread prediction**, and **wildfire** … to **wildfire** detection (A), 23.8% to **wildfire spread prediction** (B)…

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Wildfire spreading **prediction** using multimodal data and deep neural network approach

D Shadrin, S Illarionova, F Gubanov, K Evteeva… - Scientific reports, 2024 - nature.com

… critical issue in fire **behavior prediction**, so we analyze their contribution to the **model**’s results.
… In this study, our focus is on predicting **wildfire spread** using multimodal data to assist in …

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A general **model** for **wildfire propagation** with wind and slope

M Ángel Javaloyes, E Pendás-Recondo… - SIAM Journal on Applied …, 2023 - SIAM

… In the **model** we will mainly work in these coordinates, passing all the information from the
… the **wildfire propagation** more easily, and then back to \(\hat N\) to see the actual **spread**. …

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A GIS-based fire **spread** simulator integrating a simplified physical **wildland fire model** and a wind field **model**

D Prieto Herráez, MI Asensio Sevilla… - International Journal …, 2017 - Taylor & Francis

… The **wildland fire model** we propose is called Physical **Forest Fire Spread** (PhFFS) (Asensio and … It is a simplified 2D one-phase semi-physical **model** for surface fires based on the …

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Simulating **forest fire spread** and fire-fighting using cellular automata

K Mutthulakshmi, MRE Wee, YCK Wong, JW Lai… - Chinese Journal of …, 2020 - Elsevier

… of wave **propagation** allow an understanding of the natural **spread** of fire, our **model** applies
… strategies, with the spatial and **propagation dynamics** of fire considered. Cellular automata is …

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Analyzing **wildfire** evacuation **dynamics** with agent-based modeling in damaged road networks

F Ma, JY Lee - Safety Science, 2025 - Elsevier

… In this context, **wildfire** evacuation **simulation** can serve as an … **wildfire** evacuations in damaged transportation settings. The proposed framework uniquely integrates **wildfire simulation** …

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Simulating **forest fire spread** with cellular automation driven by a LSTM based speed **model**

X Li, M Zhang, S Zhang, J Liu, S Sun, T Hu, L Sun - Fire, 2022 - mdpi.com

… **forest fire spread** are studied with the accurate **prediction** of ROS and excellent performance of the **propagation** … , and CA is used as the fire **spread model**. The previous wind speed and …

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Systematic **review** and research gaps on **wildfire** evacuations: infrastructure, transportation modes, networks, and planning

SN Zehra, SD Wong - Transportation Planning and Technology, 2024 - Taylor & Francis

… We chose against searching for and including papers on simulations and **behavior** in **wildfire**

… For instance, a 2009 Australian **bushfire** case study applied a rerouting problem **model** to …

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Deciphering the impact of uncertainty on the accuracy of large **wildfire spread** simulations

A Benali, AR Ervilha, ACL Sá, PM Fernandes… - Science of the Total …, 2016 - Elsevier

… assignment and typology, location and timing of ignitions, had a major impact on **prediction**

… fire **spread simulation** approaches, and provide the necessary data for any fire **model** user …

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